

Lucas Y. Tian, Ph.D.

NINDS K99 Postdoctoral Associate
 Laboratory of Neural Systems
 The Rockefeller University
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EDUCATION

2012 - 2019 **Ph.D.** in Neuroscience, UC San Francisco, CA (Advisor: Michael Brainard)
 2007 - 2011 **B.A.** with Honors in Biology, specializing in Neuroscience, University of Chicago, IL

RESEARCH

2019 - Present **Postdoctoral Scholar**, The Rockefeller University, New York, NY
 Advisor: Winrich Freiwald; Co-advisors: Josh Tenenbaum, Xiao-Jing Wang
 Topic: Neural substrates of compositional generalization, in a macaque drawing task.

2019 (May-Dec) **Visiting Postdoctoral Scholar**, MIT, Cambridge, MA
 Advisor: Josh Tenenbaum; Co-advisors: Winrich Freiwald, Xiao-Jing Wang
 Topic: Cognitive computational modeling of human drawing.

2012 - 2019 **Graduate Student**, UCSF, San Francisco, CA
 Advisor: Michael Brainard
 Thesis: Neural mechanisms of motor skill flexibility in songbirds ([link](#))

AWARDS AND HONORS

General

2025 MSN Extramural Postdoctoral Invited Speaker, Mt. Sinai
 2025 SPINES Extramural Postdoctoral Invited Speaker, NYU
 2025 NEUROYES Extramural Postdoctoral Invited Speaker, University of Rochester
 2023 NINDS K99/R00 Transition to Independence Award
 2020 NIH BRAIN F32 NRSA Postdoctoral Fellowship
 2014 Herbert W. Boyer PIBS Fellowship, UCSF

Travel grants

2025 Postdoctoral Association Career Development Award, Rockefeller
 2025 COSYNE Presenters Travel Grant
 2016 Graduate Division Travel Award, UCSF

PEER-REVIEWED PUBLICATIONS

Tian LY, Garzón KU, Hanuska DJ, Rouse AG, Eldridge MA, Schieber MH, Wang XJ, Tenenbaum JB, Freiwald WA (2025). Neural representation of action symbols in primate frontal cortex. *Nature* (accepted in principle).

Preprint: [bioRxiv 2025.03.03.641276](https://doi.org/10.1101/2025.03.03.641276). Presentations: SfN 2024 poster ([link](#)), Cosyne 2025 talk ([link](#)).

Tian LY, Warren TL, Mehaffey WH, Brainard MS (2023). Dynamic top-down biasing implements rapid adaptive changes to individual movements. *eLife* 12:e83223. [Link](#)

Veit L, **Tian LY**, Monroy Hernandez CJ, Brainard MS (2021). Songbirds can learn flexible contextual control over syllable sequencing. *eLife* 10: e61610. [Link](#)

Tian LY, Ellis K, Kryven M, Tenenbaum JB (2020). Learning abstract structure for drawing by efficient motor program induction. *Advances in Neural Information Processing Systems*, 33 [oral presentation]. [Link](#)

Tian LY, Brainard MS (2017). Discrete circuits support generalized versus context-specific vocal learning in the songbird. *Neuron* 96, 1-10. [Link](#), [Commentary](#), [News](#)

MANUSCRIPTS IN PREPARATION

Tian LY, Hanuska DJ, Garzón KU, Liu Y, Tenenbaum JB, Wang XJ, Freiwald WA. A structured population code for a symbolic action grammar.

FUNDING

2023 - Present	NINDS K99/R00 Transition to Independence Award <i>Title: Investigating Symbolic Computation in the Brain: Neural Mechanisms of Compositionality (5K99NS131585) (PI: Tian, LY)</i> <i>Total Award Amount (including Indirect Costs): \$268,542 (2-year K99 phase), \$747,000 (3-year R00 phase, estimated, if activated before July 1, 2026)</i>
2021 - Present	Simons Foundation SCGB Pilot Award (key personnel – contributing co-author) <i>Title: Investigating Symbol-like Reasoning in the Brain: Neural Mechanisms of Compositional Action Planning (876120SPI) (PI: Freiwald, WA)</i> <i>Total Award Amount (including Indirect Costs): \$770,000</i>
2020 - 2023	NIH BRAIN F32 NRSA Postdoctoral Fellowship <i>Title: The planning of new compositional action sequences guided by interpretation of ambiguous sensory data in a novel drawing task (F32MH125573) (PI: Tian, LY)</i> <i>Total Award Amount: \$209,778</i>

INVITED TALKS

Competitively selected conference talks

2025	Society for Neuroscience Nanosymposium, San Diego, USA
2025	COSYNE Main Meeting (3% of submissions), Montreal, Canada (Recording)
2024	Ascona Circuits Meeting, Ascona, Switzerland (Slides)
2020	NeurIPS (1% of submissions) (Recording)

Symposia

2025	Kavli Internal Symposium, Rockefeller, New York, USA
2025	Lipschultz Symposium, Icahn School of Medicine (Mt. Sinai), New York, USA
2025	The Social Brain Symposium, Rockefeller and Columbia, New York, USA
2021	Object Cognition Workshop, Yale, New Haven, USA (virtual)
2018	Birds Bats Brains Meeting, UC Berkeley, USA

Departmental talks

2026	SPiNES Extramural Postdoctoral Invited Speaker, NYU, USA
2026	NEUROYES Extramural Postdoctoral Invited Speaker, University of Rochester, USA

2025	MSN Extramural Postdoctoral Invited Speaker, Mt. Sinai, USA
2025	Department of Neurobiology, University of Pittsburgh, USA
2025	German Primate Center, Leibniz Institute for Primate Research, Göttingen, Germany
2023	Kavli Internal Seminar series, Rockefeller, New York, USA
2021	Center for Brains, Minds & Machines, MIT, Cambridge, USA (Link to recording)
2021	PDA Summer Seminar Series, Rockefeller, New York, USA
2016	Asilomar Neuroscience Retreat, UC San Francisco, USA

Lab talks

2025	UCSD, Bria Long
2025	Stanford, Andreas Tolias (virtual)
2025	Western University, Jörn Diedrichsen, Paul Gribble, and Andrew Pruszynski (virtual)
2025	Netherlands Institute for Neuroscience, Pieter Roelfsema
2025	Princeton, Tim Buschman
2025	NYU, Marcelo Mattar
2025	MIT, Ev Fedorenko
2025	Newcastle University, Stuart Baker (virtual)
2025	NYU, Michael Long
2022	UCSF, Michael Brainard
2021	Yale, Ilker Yildirim (virtual)
2020	NYU, Brenden Lake and Todd Gureckis (virtual)
2020	MIT, Laura Schulz (virtual)
2016	UCSF, John Houde

PROFESSIONAL SERVICE

Workshop organization

2026	Co-organizer (w/ Mathias Sablé-Meyer), COSYNE Workshop, “Symbols as foundational to the biological basis of intelligent behavior”
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Other service

2023 - Present	Peer-reviewed for: eLife, Cognitive Science Society, Cognitive Computational Neuroscience
2014 - 2019	Co-organizer, Systems Neuroscience Research in Progress Seminar Series, UCSF
2013 - 2019	Co-organizer, Special Topics Seminar Series, UCSF

SELECT CONFERENCE ABSTRACTS

2025	Cognitive Computational Neuroscience, Amsterdam, The Netherlands
2025	Society for the Neural Control of Movement, Panama City, Panama
2024	Society for Neuroscience, Chicago, IL (Link to poster)
2024	Ascona Circuits Meeting, Ascona, Switzerland

MENTORSHIP

Research assistants

2024 - Present	Daniel Hanuska, research assistant, Rockefeller
2022 - 2024	Kedar Garzón Gupta, research assistant, Rockefeller (Current position: Ph.D. Student at Columbia)

Ph.D. rotation students

2024	Siddhartha Sharma, Freiwald lab, Rockefeller
2021	Yanis Tazi, Freiwald lab, Rockefeller
2018	Eszter Kish, Brainard lab, UCSF
2014	Rachel Care, Brainard lab, UCSF

Interns and other mentorship

2024 - 2025	Valerie Calligy, Shenoy Undergraduate Research Fellow (SURFiN), Simons Foundation, Rockefeller
2024 - 2025	Xuan Ma, Bard-Rockefeller intern, Rockefeller
2024	Daniel Dolnik, intern, Cornell Tech Master's Program
2024	Thea Wu, summer intern (Summer Undergraduate Research Fellowship, SURF), Rockefeller
2022	Alejandra Urquieta, summer intern, SURF, Rockefeller
2021	Sam Coolsaet, medical intern, Rockefeller
2021	Alex Gu, summer intern, SURF, MIT
2017	Christian Jose Monroy Hernandez, HHMI Exceptional Research Opportunities Program (EXROP), UCSF (co-mentor: Lena Veit)

OUTREACH

2024 - 2025	Mentor, Simons Foundation SURFiN, Rockefeller <i>Mentee: Valerie Calligy</i>
2021	Mentor, Neuromatch Academy (virtual) <i>Mentees: Pulido Arias, Lourdes Baztan Bitopoulos, Joseph Gonzalez, May Xia, Mark Zarutin</i>
2017	Mentor, HHMI EXROP, UCSF <i>Mentee: Christian Jose Monroy Hernandez (co-mentor: Lena Veit)</i>
2014	Judge, Synopsys Science and Technology Championship, San Jose, CA
2013	Invited speaker, PITCH, Center for Education Partnerships (CEP), UCSF
2013 - 2014	Mentor, Fairposium at Burton High School, CEP, UCSF

TEACHING

2014	TA, BMS117: Infection & Host Response (Neurophysiology unit), UCSF
2012 - 2013	Scientist-Teacher, Raul Wallenberg High School, Science and Health Education Partnership, UCSF
2012	English Teacher, San Gabriel Kindergarten, Los Nogales, Peru
2010	Teacher, Neuroscience of Illusions, Cascade!, University of Chicago

ADVANCED COURSES

2019	Methods in Computational Neuroscience, Marine Biological Laboratory, MA
2018	Jr. Workshop on Mechanistic Cognitive Neuroscience, Janelia Research Campus, VA
2018	Brains, Minds and Machines, Marine Biological Laboratory, MA
2017	Mining and Modeling of Neuroscience Data, Redwood Center, UC Berkeley, CA

REFERENCES

Winrich Freiwald, Ph.D.

Denise A. and Eugene W. Chinery Professor
Laboratory of Neural Systems, Rockefeller University
1230 York Avenue, New York, NY 10065

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Relationship: Postdoctoral advisor

Michael Brainard, Ph.D.

Professor of Physiology and Psychiatry
Departments of Physiology and Psychiatry, UCSF/HHMI
675 Nelson Rising Lane, Room 514E, San Francisco, CA 94158

michael.brainard@ucsf.edu

(415) 502-7344

Relationship: Ph.D. advisor

Joshua Tenenbaum, Ph.D.

Professor of Computational Cognitive Science
Department of Brain and Cognitive Sciences, MIT
Building 46-4015, 77 Massachusetts Avenue, Cambridge, MA 02139

jbt@mit.edu

(617) 452-2010

Relationship: Postdoctoral co-advisor

Xiao-Jing Wang, Ph.D.

Professor of Neural Science
Center for Neural Science, NYU
4 Washington Place, Room 752, New York, NY 10003

xjwang@nyu.edu

(212) 998-3677

Relationship: Postdoctoral co-advisor